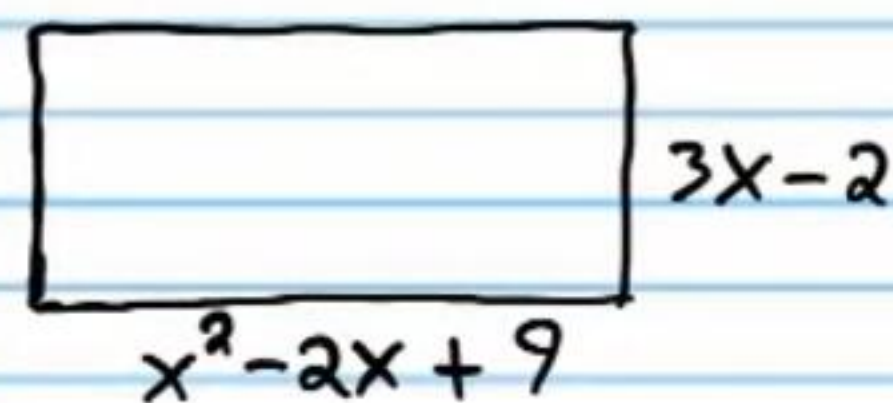


Applications of Algebra (Part II)

Name
Date
Course

① The perimeter of the rectangle below is 98 inches. Find the length of each side.



$$P = S_1 + S_2 + S_3 + S_4$$

$$98 = 3x-2 + 3x-2 + x^2-2x+9 + x^2-2x+9$$

$$98 = 2x^2 + 2x + 14$$

$$0 = 2x^2 + 2x - 84$$

$$\frac{0}{2} = \frac{2x^2 + 2x - 84}{2}$$

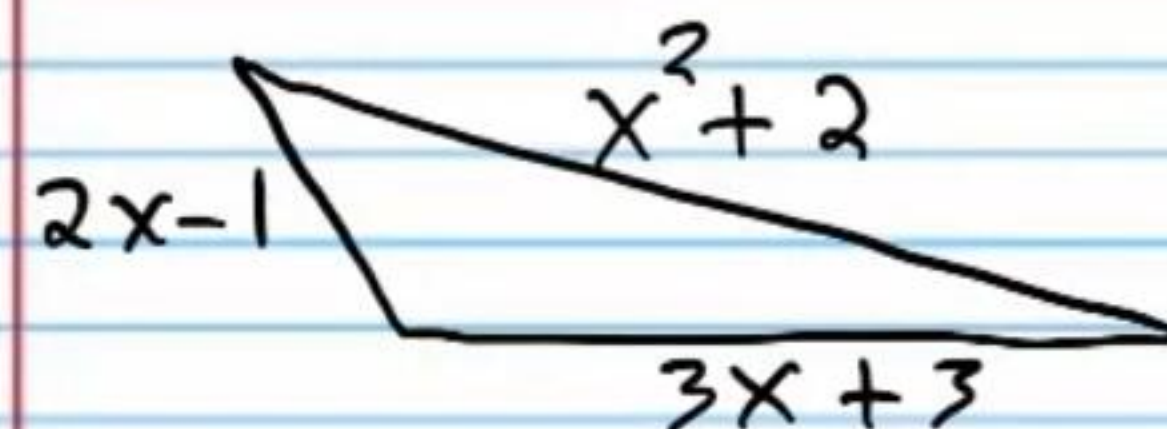
$$0 = x^2 + x - 42$$

$$0 = (x+7)(x-6)$$

$$x \neq -7, x = 6$$

$x=6$ ↓	$x=6$ ↓
$3x-2$	x^2-2x+9
$3(6)-2$	$6^2-2(6)+9$
$18-2$	$36-12+9$
16 in	33 in

② The perimeter of the triangle below is 40 feet. Find the length of each side.



$$P = S_1 + S_2 + S_3$$

$$40 = 2x-1 + 3x+3 + x^2+2$$

$$40 = x^2 + 5x + 4$$

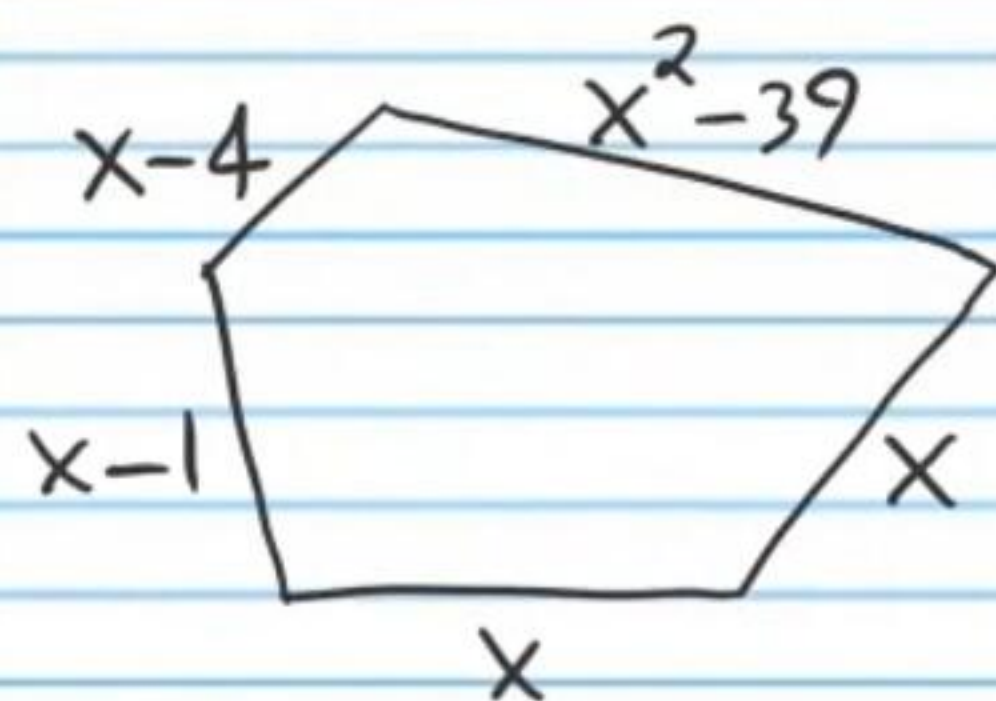
$$0 = x^2 + 5x - 36$$

$$0 = (x+9)(x-4)$$

$$x \neq -9, x = 4$$

$x=4$ ↓	$x=4$ ↓	$x=4$ ↓
$2x-1$	$3x+3$	x^2+2
$2(4)-1$	$3(4)+3$	4^2+2
$8-1$	$12+3$	$16+2$
7 ft	15 ft	18 ft

③ The perimeter of the pentagon below is 33 feet. Find the length of each side.



$$P = S_1 + S_2 + S_3 + S_4 + S_5$$

$$33 = x-4 + x-1 + x + x + x^2-39$$

$$33 = x^2 + 4x - 44$$

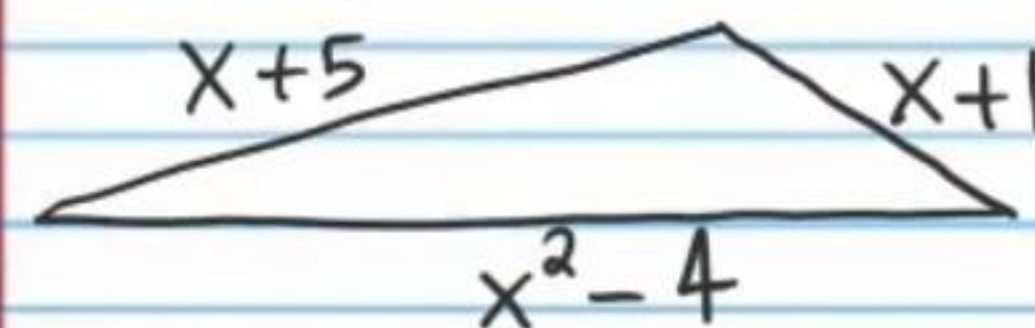
$$0 = x^2 + 4x - 77$$

$$0 = (x+11)(x-7)$$

$$x \neq -11, x = 7$$

$x=7$	$x=7$	$x=7$	$x=7$	$x=7$
$\Downarrow$	$\Downarrow$	$\Downarrow$	$\Downarrow$	$\Downarrow$
x	x	$x-1$	$x-4$	x^2-39
$\boxed{7 \text{ ft}}$	$\boxed{7 \text{ ft}}$	$7-1$	$7-4$	7^2-39
		$\boxed{6 \text{ ft}}$	$\boxed{3 \text{ ft}}$	$49-39$
				$\boxed{10 \text{ ft}}$

④ The perimeter of the triangle below is 26 meters. Find the length of each side.



$$P = S_1 + S_2 + S_3$$

$$26 = x^2 - 4 + x + 5 + x + 1$$

$$26 = x^2 + 2x + 2$$

$$0 = x^2 + 2x - 24$$

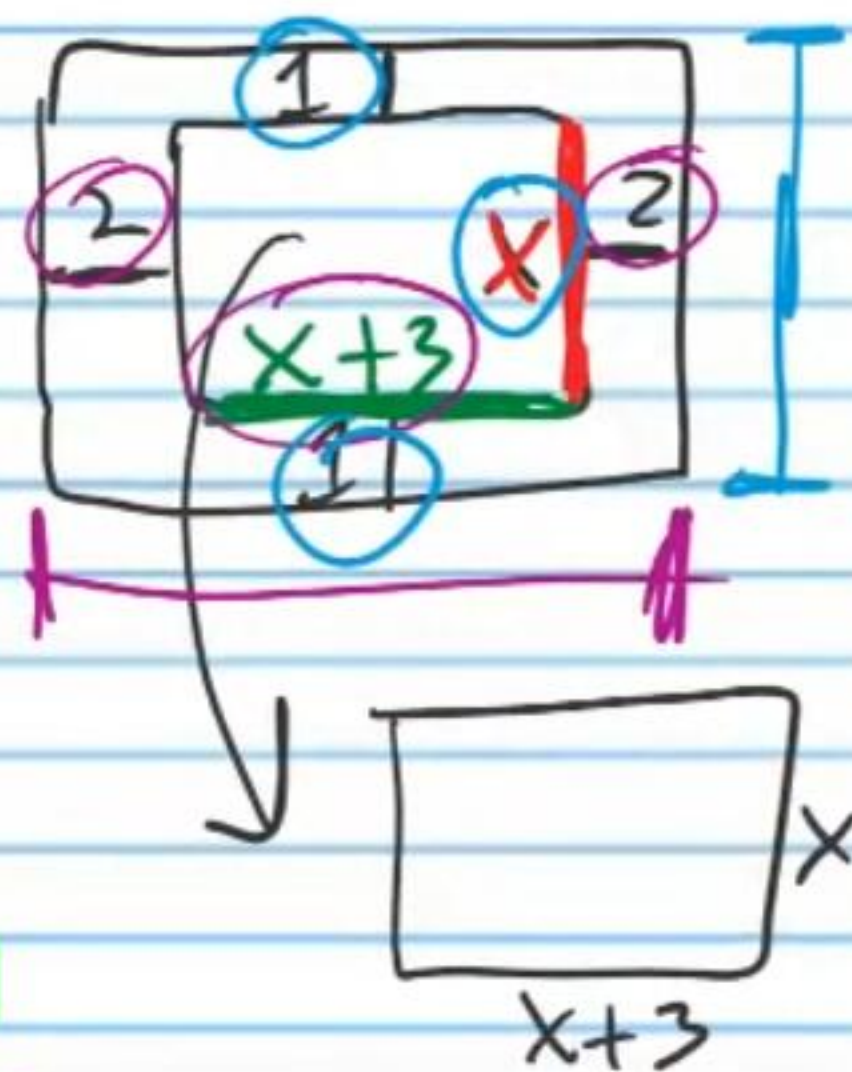
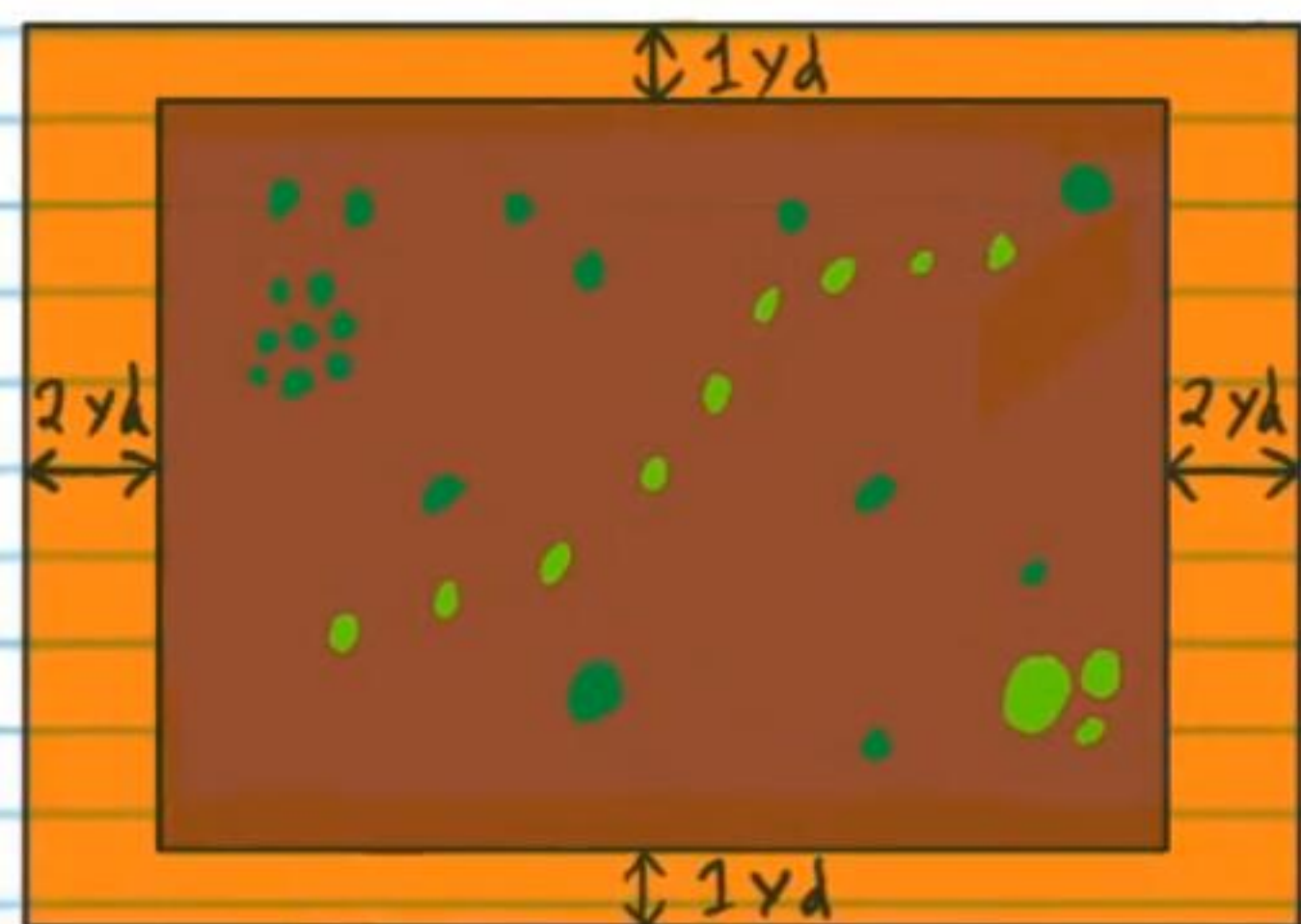
$$0 = (x+6)(x-4)$$

$$x \neq -6, x = 4$$

$x=4$	$x=4$	$x=4$
$\Downarrow$	$\Downarrow$	$\Downarrow$
$x+5$	$x+1$	x^2-4
$4+5$	$4+1$	4^2-4
$\boxed{9 \text{ m}}$	$\boxed{5 \text{ m}}$	$16-4$
		$\boxed{12 \text{ m}}$

- ⑤ A rectangular garden contains an inner rectangle with soil and plants and an outer walkway. The width of the inner rectangle is three more yards than the length and the area of the entire garden is 134 more square yards than the area of the inner part. What are the dimensions of the garden?

★ Not drawn to scale.



$$A_G = 134 + A_I$$

$$(x+3+2+2)(x+1+1) = 134 + x(x+3)$$

$$(x+7)(x+2) = 134 + x(x+3)$$

$$x^2 + 2x + 7x + 14 = 134 + x^2 + 3x$$

$$x^2 + 9x + 14 = 134 + x^2 + 3x$$

$$9x + 14 = 134 + 3x$$

$$6x = 120$$

$$x = 20$$

$$w = x + 7$$

$$w = 20 + 7$$

$$w = 27 \text{ yd}$$

$$x = 20$$

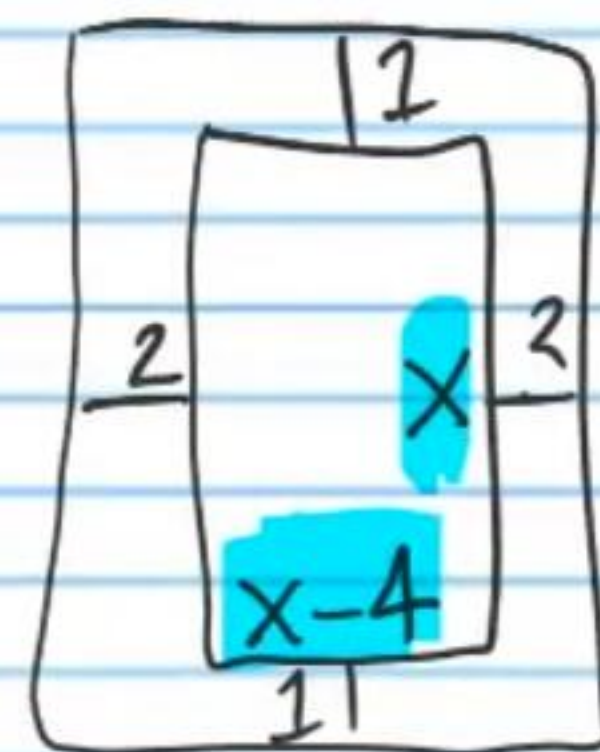
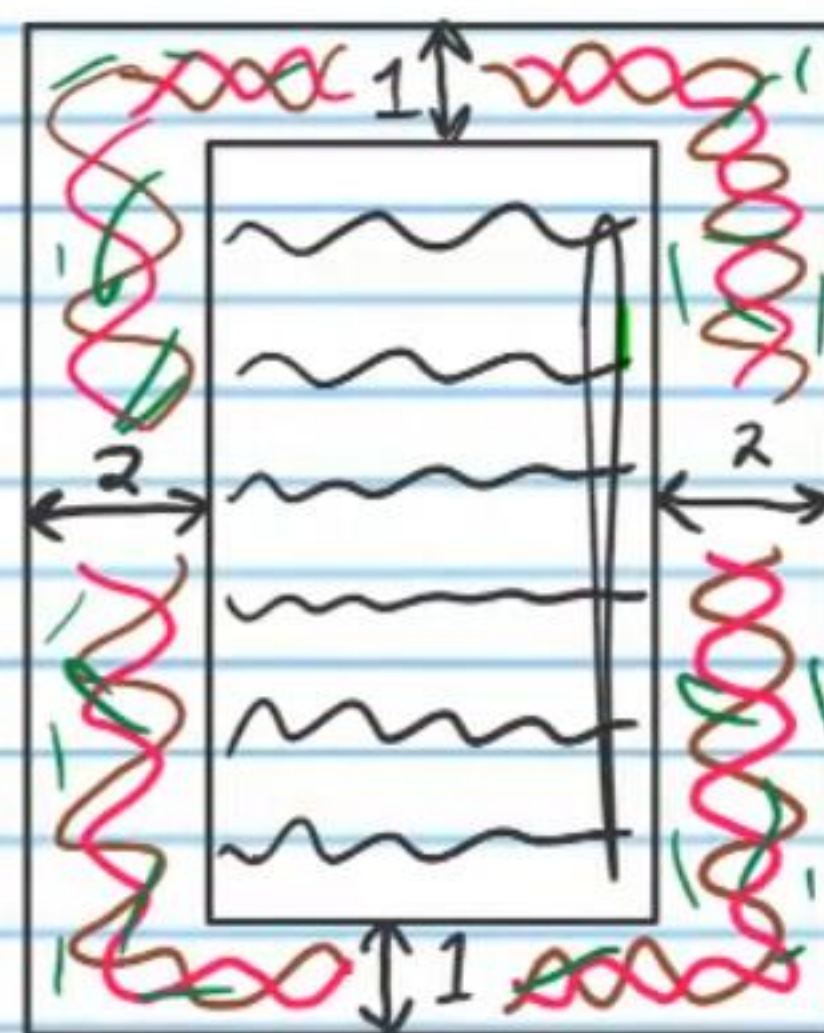
$$L = x + 2$$

$$L = 20 + 2$$

$$L = 22 \text{ yd}$$

- ⑥ A rectangular wedding invitation has an outer border that is decorated and an inner rectangular portion with text. The width of the inner portion is four inches less than the length and the area of the entire invitation is three times the area of the inner portion with text. Find the dimensions of the invitation.

★ Not drawn to scale.



$$A_I = 3A_T$$

$$(x+1+1)(x-4+2+2) = 3[x(x-4)]$$

$$x = 7$$

$$L = x + 2$$

$$L = 7 + 2$$

$$L = 9 \text{ in}$$

$$(x+2)(x) = 3x(x-4)$$

$$x^2 + 2x = 3x^2 - 12x$$

$$0 = 2x^2 - 14x$$

$$0 = 2x(x-7)$$

$$x \neq 0$$

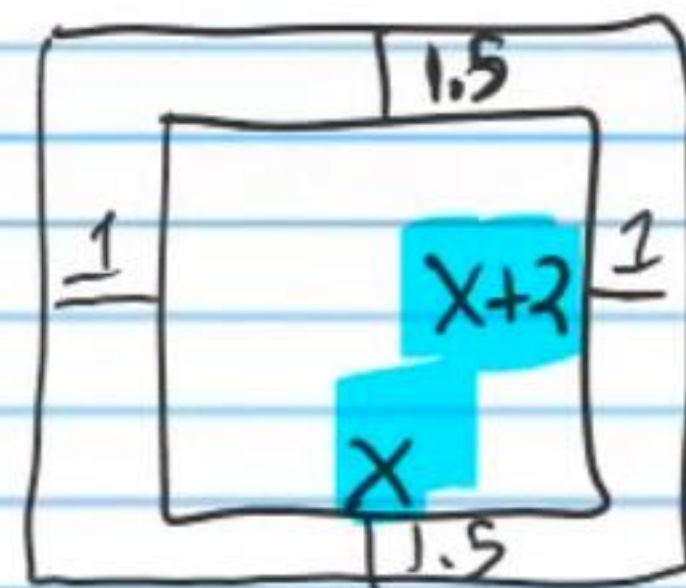
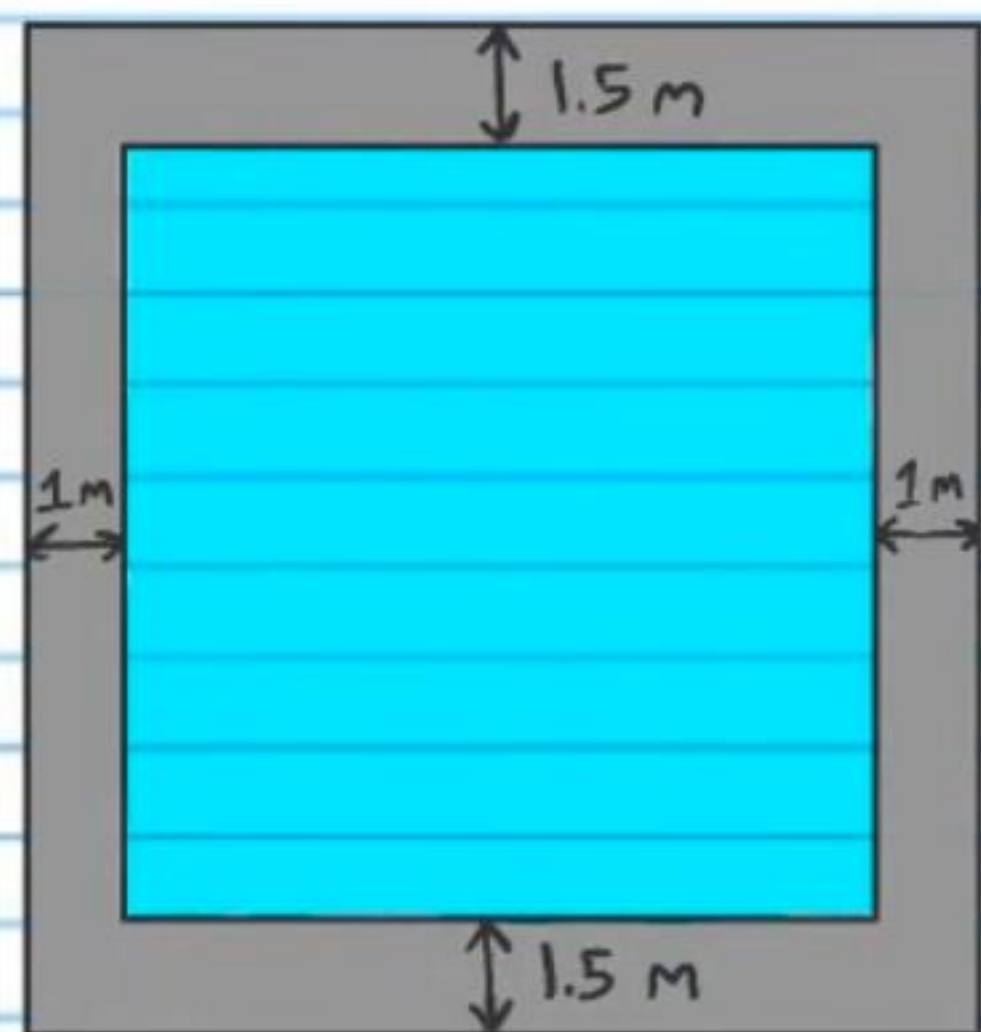
$$x = 7$$

$$w = x$$

$$w = 7 \text{ in}$$

⑦ The total area of a rectangular backyard is 50 square meters more than the area of the rectangular pool at the center of the yard. If the length of the pool is two meters more than the width, what are the dimensions of the backyard and pool?

★ Not drawn to scale.



$$A_B = 50 + A_P$$

$$(x+1+1)(x+2+1.5+1.5) = 50 + x(x+2)$$

$$(x+2)(x+5) = 50 + x(x+2)$$

$$x^2 + 7x + 10 = 50 + x^2 + 2x$$

$$7x + 10 = 50 + 2x$$

$$5x = 40$$

$$x = 8$$

$$L = 10 \text{ m}, W = 8 \text{ m}$$

$$x = 8$$

$$W = x + 2$$

$$W = 8 + 2$$

$$W = 10 \text{ m}$$

$$x = 8$$

$$L = x + 5$$

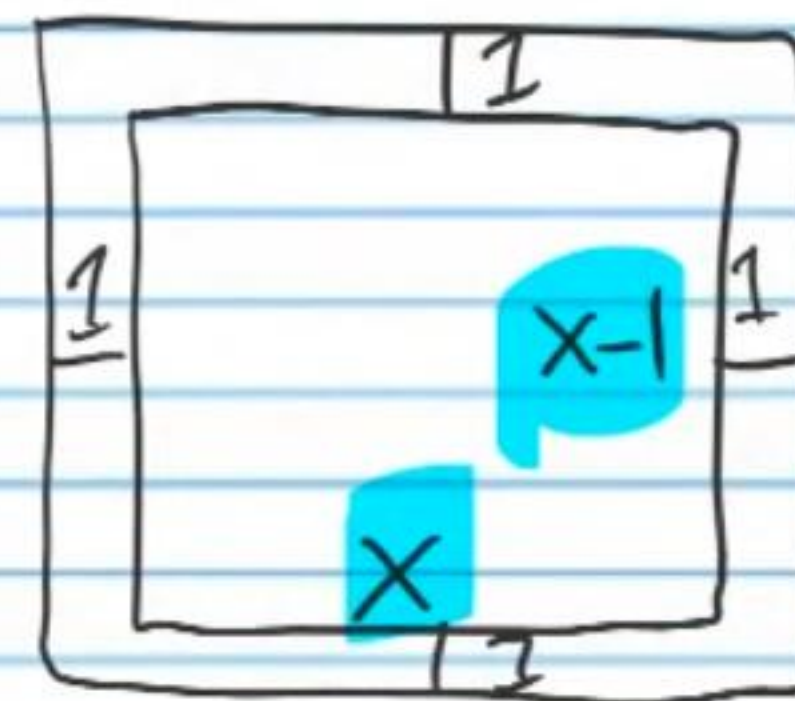
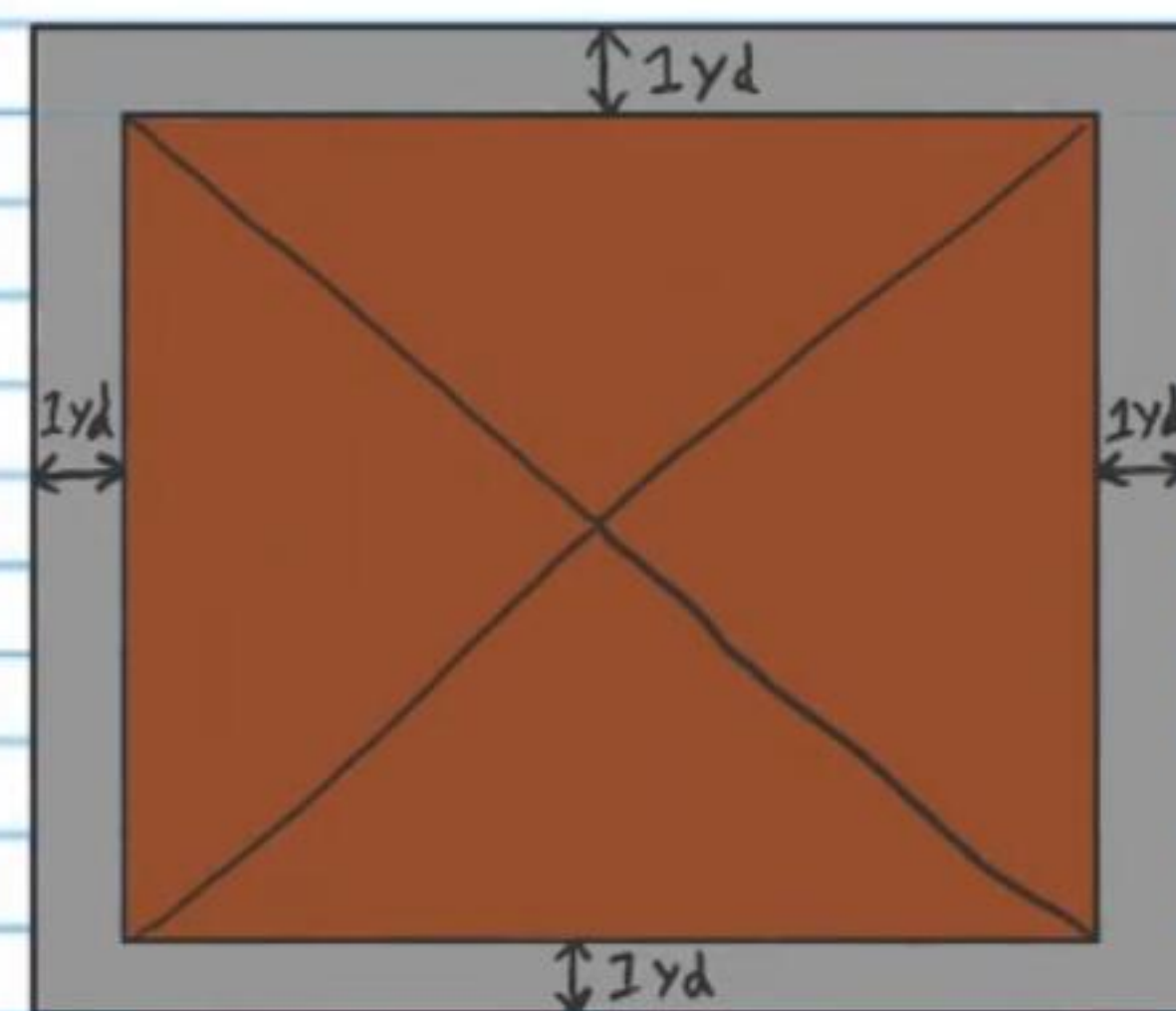
$$L = 8 + 5$$

$$L = 13 \text{ m}$$

Combined

⑧ The base of a house is a rectangle. There is a concrete walkway around the perimeter of the house. The length of the house is one yard less than the width. The area of the base of the house and the walkway combined is 64 square yards less than twice the area of the base of the house. What are the dimensions of the house?

★ Not drawn to scale.



$$A_c = 2A_B - 64$$

$$(x+1+1)(x-1+1+1) = 2[x(x-1)] - 64$$

$$(x+2)(x+1) = 2[x^2 - x] - 64$$

$$x^2 + 3x + 2 = 2x^2 - 2x - 64$$

$$0 = x^2 - 5x - 66$$

$$0 = (x-11)(x+6)$$

$$x = 11, x \neq -6$$

$$x = 11$$

$$W = x$$

$$W = 11 \text{ yd}$$

$$x = 11$$

$$L = x - 1$$

$$L = 11 - 1$$

$$L = 10 \text{ yd}$$

⑨ After flying for a few moments, a toy rocket starts to descend. If we neglect air resistance, the height (in feet) of the rocket after t seconds of free fall can be modeled by the following formula.

$$h = -16t^2 + 200$$

a) Find the height of the rocket three seconds after free fall. b) How many seconds after free fall will pass before the rocket is 80 feet above the ground? c) 100 feet? d) When will the rocket hit the ground?

$$a) h = -16(3)^2 + 200 = -16(9) + 200 = -144 + 200 = \boxed{56 \text{ ft}}$$

$$b) 80 = -16t^2 + 200 \quad c) 100 = -16t^2 + 200$$

$$-120 = -16t^2$$

$$\frac{-120}{-16} = t^2$$

$$\frac{15}{2} = t^2$$

$$\pm \sqrt{\frac{15}{2}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{15}{2}} = t$$

$$\pm \frac{\sqrt{15}}{\sqrt{2}} = t$$

$$\pm \frac{\sqrt{15} \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = t$$

$$\pm \frac{\sqrt{30}}{2} = t$$

$$-100 = -16t^2$$

$$\frac{-100}{-16} = t^2$$

$$\frac{25}{4} = t^2$$

$$\pm \sqrt{\frac{25}{4}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{25}{4}} = t$$

$$\pm \frac{5}{2} = t$$

$$\boxed{\frac{5}{2} \text{ sec} = t}$$

$$d) 0 = -16t^2 + 200$$

$$-200 = -16t^2$$

$$\frac{-200}{-16} = t^2$$

$$\frac{25}{2} = t^2$$

$$\pm \sqrt{\frac{25}{2}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{25}{2}} = t$$

$$\pm \frac{5}{\sqrt{2}} = t$$

$$\pm \frac{5 \cdot \sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = t$$

$$\pm \frac{5\sqrt{2}}{2} = t$$

$$\boxed{\frac{5\sqrt{2}}{2} \text{ sec} = t}$$

⑩ A chipmunk drops an acorn from a tall tree. Neglecting air resistance, the height (in feet) of the acorn after t seconds in the air can be modeled by the formula below.

$$h = -16t^2 + 73$$

a) What is the height of the acorn after two seconds in the air? b) How long will it take for the acorn's height to reach 60 feet? c) 35 feet? d) How long will it take for the acorn to hit the ground?

$$a) h = -16(2)^2 + 73 = -16(4) + 73 = -64 + 73 = \boxed{9 \text{ ft}}$$

$$b) 60 = -16t^2 + 73 \quad c) 35 = -16t^2 + 73 \quad d) 0 = -16t^2 + 73$$

$$-13 = -16t^2$$

$$\frac{13}{16} = t^2$$

$$\pm \sqrt{\frac{13}{16}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{13}{16}} = t$$

$$\pm \frac{\sqrt{13}}{4} = t$$

$$\boxed{\frac{\sqrt{13}}{4} \text{ sec} = t}$$

$$-38 = -16t^2$$

$$\frac{19}{8} = t^2$$

$$\pm \sqrt{\frac{19}{8}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{19}{8}} = t$$

$$\pm \frac{\sqrt{19}}{\sqrt{8}} = t$$

$$\pm \frac{\sqrt{19}}{\sqrt{4 \cdot 2}} = t$$

$$\pm \frac{\sqrt{19}}{2\sqrt{2}} = t$$

$$\pm \frac{\sqrt{19} \cdot \sqrt{2}}{2\sqrt{2} \cdot \sqrt{2}} = t$$

$$\pm \frac{\sqrt{38}}{4} = t$$

$$\boxed{\frac{\sqrt{38}}{4} \text{ sec} = t}$$

$$-73 = -16t^2$$

$$\frac{73}{16} = t^2$$

$$\pm \sqrt{\frac{73}{16}} = \sqrt{t^2}$$

$$\pm \sqrt{\frac{73}{16}} = t$$

$$\pm \frac{\sqrt{73}}{\sqrt{16}} = t$$

$$\pm \frac{\sqrt{73}}{4} = t$$

$$\boxed{\frac{\sqrt{73}}{4} \text{ sec} = t}$$

$$\sqrt{ab} = \sqrt{a} \cdot \sqrt{b}$$

⑪ A boy kicks a soccer ball into the air. Neglecting air resistance, the height (in feet) of the ball t seconds after it is kicked can be modeled by the equation below.

$$h = -16t^2 + 10t$$

a) How high is the ball $\frac{3}{5}$ of a second after it is kicked?

b) When will the ball be one foot in the air (hint: there are two answers)?

c) How long will it take for the ball to hit the ground (Hint: there are two answers—one is useful and the other is not)?

$$a) h = -16\left(\frac{3}{5}\right)^2 + 10\left(\frac{3}{5}\right) = -16\left(\frac{9}{25}\right) + \frac{10}{1}\left(\frac{3}{5}\right) = -16\left(\frac{4}{25}\right) + \frac{20}{5}$$

$$b) 1 = -16t^2 + 10t$$

$$0 = -16t^2 + 10t - 1$$

$$0 = -16t^2 + 10t - 1$$

$$0 = 16t^2 - 10t + 1$$

$$0 = 16t^2 - 10t + 1$$

$$0 = (2t-1)(8t-1)$$

$$0 = (2t-1)(8t-1)$$

$$2t-1=0, \quad 8t-1=0$$

$$t = \frac{1}{2} \text{ sec}, \quad t = \frac{1}{8} \text{ sec}$$

$$c) 0 = -16t^2 + 10t$$

$$0 = -2t(8t-5)$$

$$t=0, \quad 8t-5=0$$

$$t = \frac{5}{8} \text{ sec}$$

$$= 1\frac{11}{25} \text{ ft}$$

⑫ A ball bounces off the ground. Neglecting air resistance, the height (in feet) of the ball t seconds after bouncing off the ground is given by the formula below.

$$h = -16t^2 + 14t$$

a) What is the height of the ball $\frac{1}{10}$ of a second after bouncing off the ground?

b) When will the ball reach a height of 3 feet (Hint: there are two answers)?

c) When will the ball hit the ground (Hint: there are two answers—one is relevant and the other is not)?

$$a) h = -16\left(\frac{1}{10}\right)^2 + 14\left(\frac{1}{10}\right) = -16\left(\frac{1}{100}\right) + \frac{14}{10} = -\frac{4}{25} + \frac{14}{10}$$

$$b) 3 = -16t^2 + 14t$$

$$0 = -16t^2 + 14t - 3$$

$$0 = 16t^2 - 14t + 3$$

$$0 = (8t-3)(2t-1)$$

$$t = \frac{3}{8} \text{ sec}, \quad t = \frac{1}{2} \text{ sec}$$

$$c) 0 = -16t^2 + 14t$$

$$0 = -2t(8t-7)$$

$$t = 0 \text{ sec}, \quad t = \frac{7}{8} \text{ sec}$$

$$= 1\frac{6}{25} \text{ ft}$$

⑬ A ball is thrown straight down from the top of a 63 foot tall building. Its height (in feet) after t seconds in the air is given by the equation below (this equation neglects air resistance).

$$h = -16t^2 - 8t + 63$$

a) What is the height of the ball after 1.2 seconds in the air?

b) How long will it take for the ball to reach a height of 55 feet?

c) How long will it take for the ball to hit the ground?

$$a) h = -16(1.2)^2 - 8(1.2) + 63 = -16(1.44) - 8(1.2) + 63$$

$$b) 55 = -16t^2 - 8t + 63 \quad -23.04 - 9.6 + 63$$

$$0 = -16t^2 - 8t + 8 \quad -32.64 + 63$$

$$0 = \frac{-16t^2 - 8t + 8}{-8} \quad \text{c) } 0 = -16t^2 - 8t + 63 \quad \boxed{30.36 \text{ ft}}$$

$$0 = 2t^2 + t - 1 \quad 0 = -16t^2 - 8t + 63$$

$$0 = (2t - 1)(t + 1) \quad 0 = 16t^2 + 8t - 63$$

$$t = \frac{1}{2} \text{ sec} \quad t \neq -1$$
$$0 = (4t + 7)(4t - 7)$$
$$t \neq -\frac{7}{4}, \quad t = \frac{7}{4} = 1\frac{3}{4} \text{ sec}$$

14 An audience member sitting in a balcony seat throws a rose to a performer. The height of the rose (in feet) after t seconds in the air is given by the equation below. The equation neglects air resistance.

$$h = -16t^2 + 4t + 12$$

a) Find the height of the rose after 0.8 seconds in the air (the stage is $h=0$).

b) If the performer jumps off a ladder and catches the rose when it is ten feet above the stage, how long was the rose in the air?

c) If the performer does not catch the rose and it lands on the stage, for how many seconds was it in the air?

$$a) h = -16(0.8)^2 + 4(0.8) + 12 = -16(0.64) + 4(0.8) + 12$$

$$b) 10 = -16t^2 + 4t + 12 \quad = -10.24 + 3.2 + 12$$

$$0 = -16t^2 + 4t + 2 \quad = \quad 4.96 \text{ ft}$$

$$0 = -16t^2 + 4t + 2$$

$$2 - 2$$

$$0 = 8t^2 - 2t - 1$$

$$0 = (4t+1)(2t-1)$$

$$t \neq -\frac{1}{4} \Rightarrow t = \frac{1}{2} \text{ sec}$$

$$= -10.24 + 3.2 + 12$$

$$= 4.96 \text{ ft}$$

$$c) 0 = -16t^2 + 4t + 12$$

$$0 = -k^2 + 1 + 12$$

~~$$\begin{array}{r} 0 = 16 + 14 + 11 \\ -4 \quad -4 \end{array}$$~~

$$0 = 4t^2 - t - 3$$

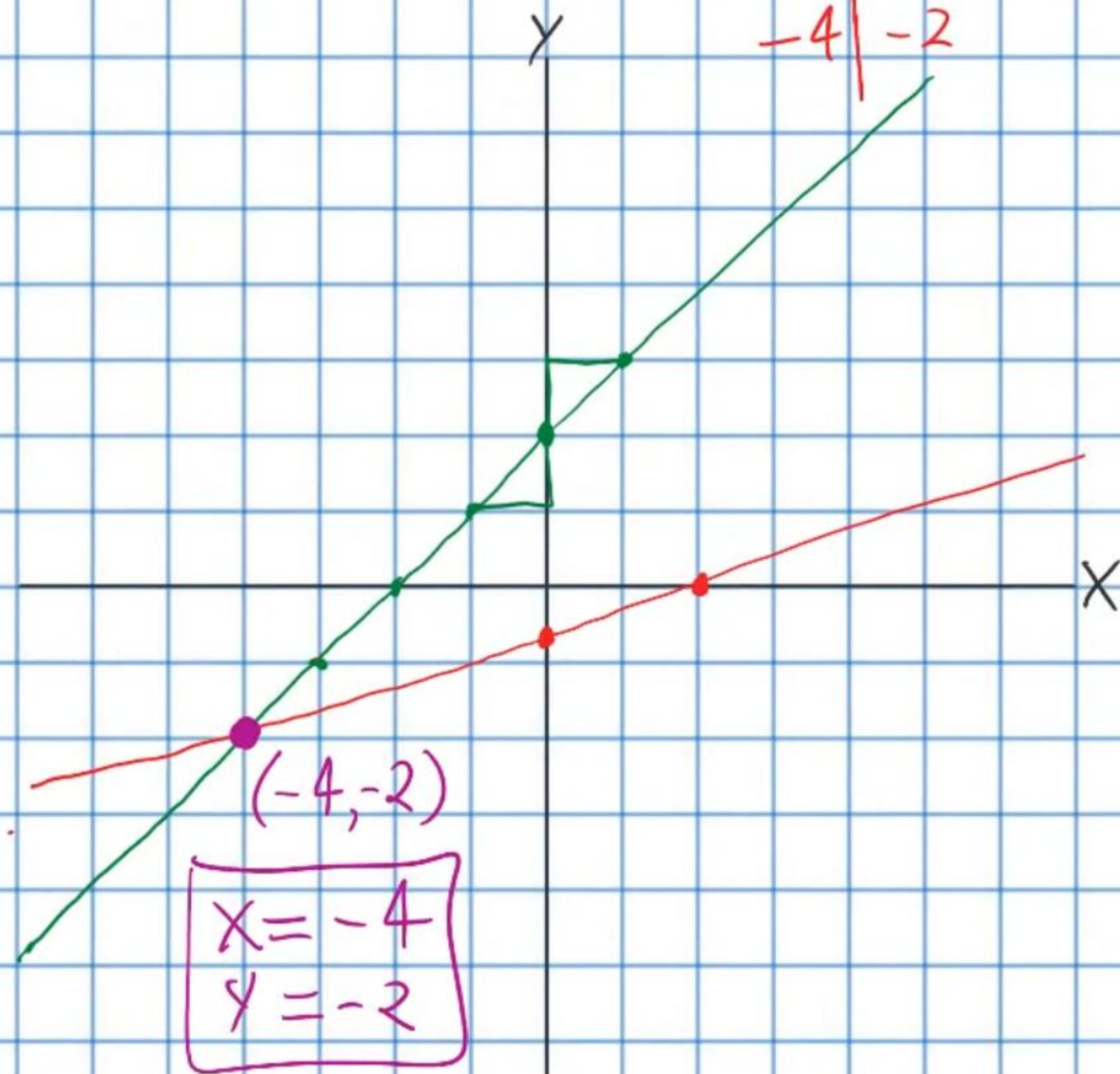
$$Q = (4+3)(t-1)$$

$$t \neq -\frac{2}{3}, \quad |t| = 1 \text{ sec}$$

Solving A System of Equations by Graphing

⑮ $\begin{cases} x-3y=2 \\ y=x+2 \end{cases}$

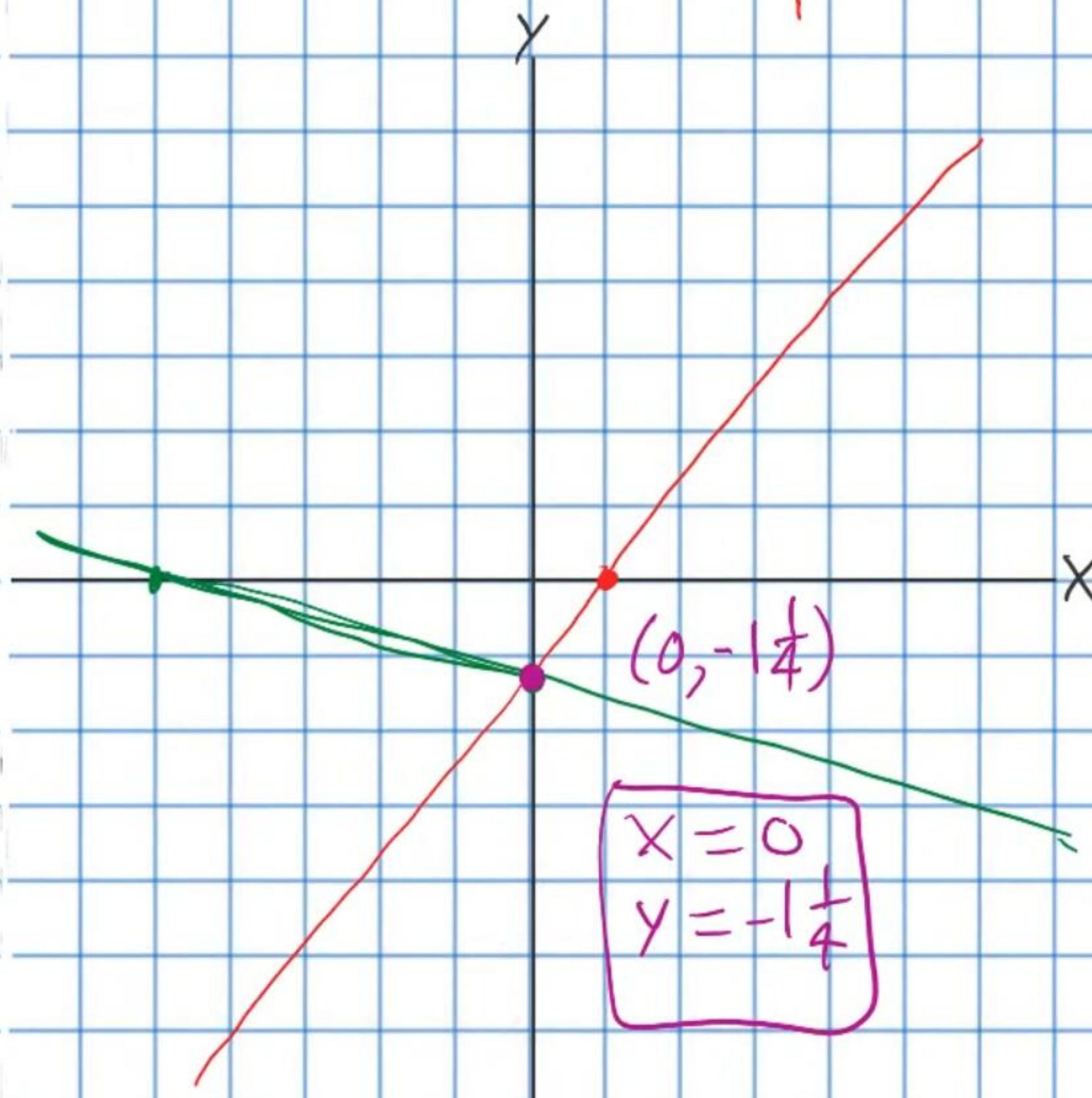
X	Y
0	$-\frac{2}{3}$
2	0
-4	-2



⑯ $\begin{cases} 5x-4y=5 \\ x+4y=-5 \end{cases}$

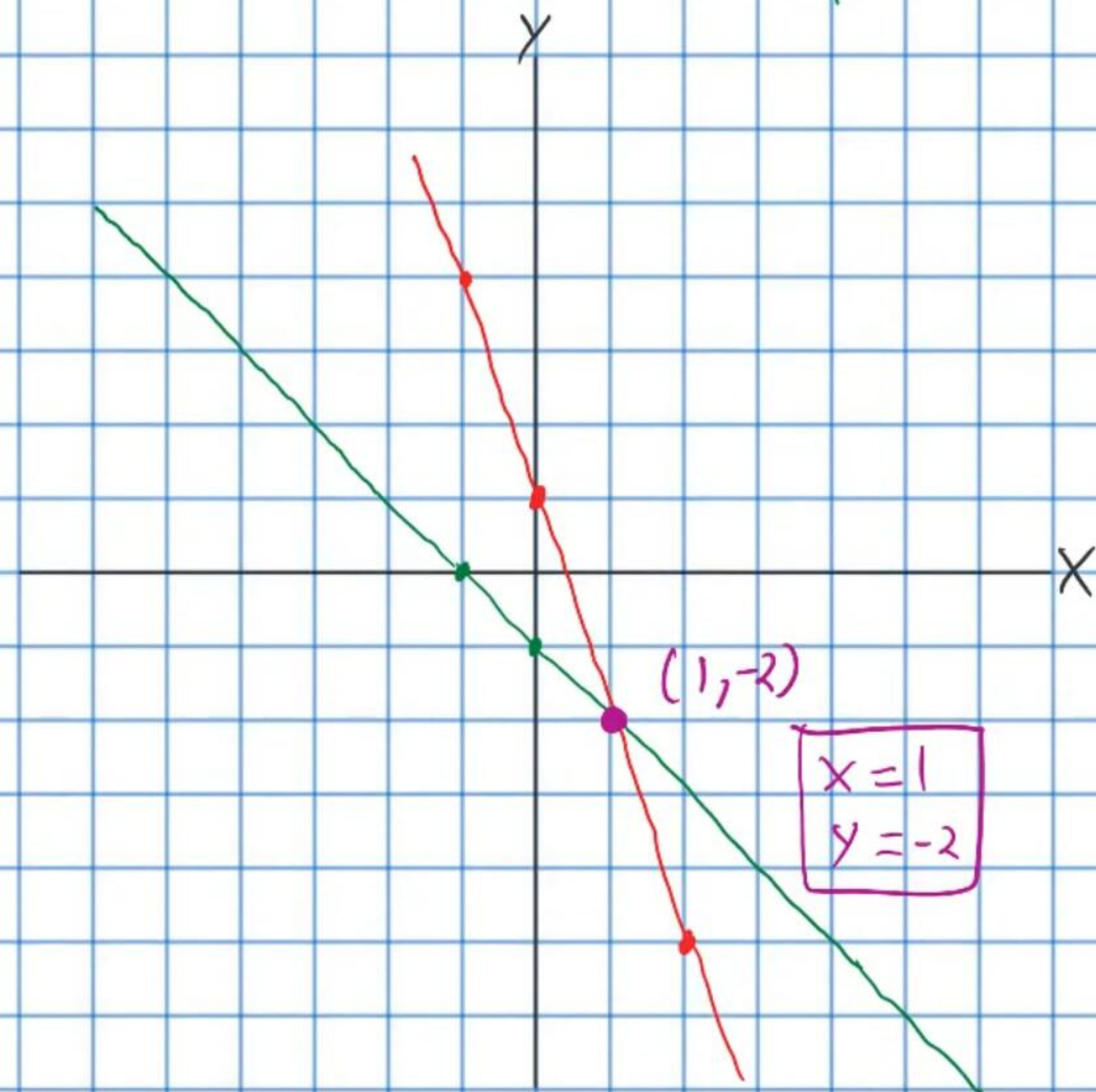
X	Y
0	$-\frac{1}{4}$
-5	0

X	Y
0	$-\frac{1}{4}$
1	0



$$\textcircled{17} \begin{cases} y = -3x + 1 \\ x + y = -1 \end{cases}$$

$$\begin{array}{c|c} x & y \\ \hline 0 & -1 \\ -1 & 0 \end{array}$$



$$\textcircled{18} \begin{cases} 2x - 4 = y \\ -6x + 2y = -9 \end{cases}$$

$$\begin{array}{c|c} x & y \\ \hline 0 & -4\frac{1}{2} \\ 1\frac{1}{2} & 0 \end{array}$$

